

MEET VADDORIYA

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WORK EXPERIENCE

Machine Learning Engineer - Product Development

Gartner Research

Feb 2024 – Present

Hyderabad, India

- Trained multi class sentence classifier for segregating sentences into different buckets in order to input the **NER** the most appropriate sentences, achieved an **ROC-AUC** score of **89%** on average for every class.
- Worked in designing and developing the **Recommender systems** for recommending the best experts who would most likely be able to solve the business problem, have explored and researched on Neural Collaborative Filtering (**NeuMF**), **Factorisation Machines**, Sequential Recommender Systems etc, for designing these systems, and achieved the highest hit-ratio of **0.7** on average for all industries like Sales, Finance etc, for the **BERT4Rec** model.
- Worked in developing and designing **Learning to Rank (LTR)** models in order to **re-rank** the experts which are best suited for the given business problems.
- Worked on different fine-tuning techniques like **PEFT, SFT, contrastive learning** etc open source LLMs like **GIST, Mistral, LLaMa** for developing more context aware models for the business.
- Deployed Expert based recommendation system using latest MLOps tools like **Weights and biases** for tracking the model training data, **Mlflow** for model versioning and artifact registry, **grafana** for api performance monitoring and auto-training on **AWS Sagemaker** which we're triggered from **AWS Lambda**, via a Cron job which had been written in **AWS EventBridge**

Machine Learning Engineer - Product Development

Phenom People

Jul 2022 – Feb 2024

Hyderabad, India

- Developed Sentence classifiers for multiple languages (**Multilingual**) achieving an average **ROC-AUC** of **80%** for classifying sentences into categories like education, experience, etc., enhancing the search functionality significantly.
- Developed a **Named-Entity-Recognition System (NER)** for extracting potential **skills** from a given job-description, this system leverages the **BERT** architecture for extracting important skills leveraging the **attention mechanism**, and achieved a precision rate of **85%** in extracting the skills.
- Developed a **Job-Parsing** service using **LLMs** for scraping out the important sentences like Experience, Education etc., from the job-description, this LLM was fine-tuned using SFT and techniques like **LORA** and **Q-LORA**, was able to achieve an **F1-Score** of **89%**.
- Developed a **Resume-parsing** service which parses resumes from texts as well as images, so we had 2 versions of the service, one which uses **Computer vision** algorithms like **Object detection**, extracts the particular text using **OCR** technique, then we pass this extracted texts to our Sentence Classification models which classifies the important sentences to undertake, then we pass them to our **NER** models for **entity extraction**, achieved an average **F1-score** of **85%** on sentence classification tasks and over **80%** average f1 score on Entity extraction tasks, so this service was modeled in python, hugging-face and pytorch and was deployed as a **microservice** in **Java**.
- Developed a **Search LTR** system for re-ranking the keyword search results, based on the frequent user-interaction, developed a reward based **Multi-armed-bandit** framework which would re-rank the search results based on rewards assigned.
- Developed a **Recommendation system** "Apply/Cart-based-recommendations" from scratch which recommends jobs based on user's previous applications/ jobs saved in the cart, this system leverages **BERT** for contextual understanding of user's and then fine tune's a **Neural Network** for providing recommendations, this system achieved a CTR of **4.5%** from the previous existing system which was based on string matching.
- Developed a **Recommendation system** "people-also-viewed" using **Deep Collaborative Filtering** which understands the user-user and user-item patterns and recommends jobs based on user preferences, this is a top-notch widget which achieved a CTR of **5%** CTR from the previous existing system which was based on Elastic Search.
- Developed a **Sequential Recommendation system** "browsing-based-recommendations" **BERT4Rec** which gives recommendations based on user's browsing history, this system was developed using concepts of **BERT** which leverages the BERT for understanding the user pattern in a session and giving recommendations based on the interactions, this system achieved an CTR of **3.5** from the existing system.
- Engineered **Recommendation systems** for making them available for all clients on large scale using **Kubernetes, Kafka and AWS Lambda**, monitored the metrics by creating dashboards on **Grafana** using grafana query language.
- Developed a **Recommendation system** which recommends jobs based on the blogs which the companies use, this system was developed using **Open-AI** and **semantic search**
- Integrated and Fine-Tuned a **Learning to Rank (LTR)** model with all the **Recommendation systems** developed, which enhanced user engagement by providing more relevant recommendations at the top, this system improved the **click rate** by **7.5%** than the normal recommendations which we're not ordered based on ranking.
- Spearheaded the integration of **Unit-testing** using pytest, robot, and coverage for all team services. Achieved **95%** test coverage and reduced critical bugs by **50%** through **Rigorous testing**

Machine Learning Intern -Product Development

Phenom People

Jul 2021 – Jul 2022

Hyderabad, India

- Developed a bias detection service for job descriptions based on the "masculinity score," utilizing **semantic similarity** and search concepts. Achieved a precision rate of **92%** in detecting biased language.
- Conducted Ad-hoc analysis on Sentence Embeddings (**XLMR and BERT**) and generated comparative reports on the performance of XLMR Embeddings for English sentence classification. Demonstrated a **15%** increase in accuracy using XLMR Embeddings over BERT.
- Created a Custom**Named-Entity-Recognition System (NER)** for skill tagging and relevant experience extraction from sentences,leading to a **25%** improvement in skill importance detection.
- Performed extensive ad-hoc analysis on Extractive and **Abstractive Summarization** models,implementing **Pagerank** and Hugging Face Transformers. Achieved a **30%** improvement in summary quality for job descriptions.

Data Science Intern

Pi-Analytix Edutech

May 2021 – Jul 2020

Remote

- Worked on Developing **Full stack Data Science** Applications and Deploying Machine Learning Pipelines on various cloud platforms like **AWS, Azure, IBM Cloud**.
- Mentored students on how to make end-to-end **data science** projects which would be solving a specific **business use-case** by explaining real time use-cases, helped students to deploy their intial release products on large scale using various **cloud platforms**.

Data Science Intern

Neubrain Solutions Pvt.Ltd.

Mar 2021 – Apr 2021

Remote

- Worked on **data collection** and data gathering aspects, using web scraping tools like **selenium, WebCrawler** and automation tools like **Octoparse**
- Developed **Analytical dashboards** on data scraped from various websites,**Rms feeds**, mobile applications etc, using javascript libraries like **D3.js**
- Developed a **Analytical dashboard** chatting application data which used to give the user's the insights of the links shared in a group at one-place, help user's understand their sentiments in the chats and also help user's to understand their interaction pattern, this dashboard was made using the concept of **Text Analytics** and **D3.js**

CERTIFICATIONS

Oracle Generative AI Professional Certification | Oracle

Jun 2024

Google Data Analytics Professional Certification | Google

Mar 2021

Google IT Automation Professional Certification | Google

Jan 2021

EDUCATION

Anurag university

Bachelor of Technology, Information Technology

Jul 2022

Hyderabad, India

SKILLS

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|-------------------------------------|-----------------------------|----------------|------------------------|
| – Data Science | – Prompt Engineering | – Pytorch | – Elastic Search |
| – Product Development | – Software Engineering | – Tensorflow | – Kubernetes |
| – Machine Learning | – Data Engineering | – MIOps | – Docker |
| – Deep Learning | – System Design (HLD, LLD) | – Numpy | – Google Suite |
| – Natural Language Processing (NLP) | – Software Testing | – OpenCV | – Microsoft Excel |
| – Recommender Systems | – Amazon Web Services (AWS) | – Pandas | – Microsoft Powerpoint |
| – Large Language Models (LLM) | – Python | – SQL | – Microsoft PowerBi |
| | – Java | – NoSQL | – JavaScript |
| | | – Apache Kafka | – WebScraping |